

Course overview

Introduction to computer programming

Management and Artificial Intelligence



Greetings, folks!

Welcome to the ***Introduction to Computer Programming***
course!

Important Resources

Web Resources you should have access to:

- **MyLUISS**
- **Course Syllabus**
- **Rooms and Timing**

e-Learning platform

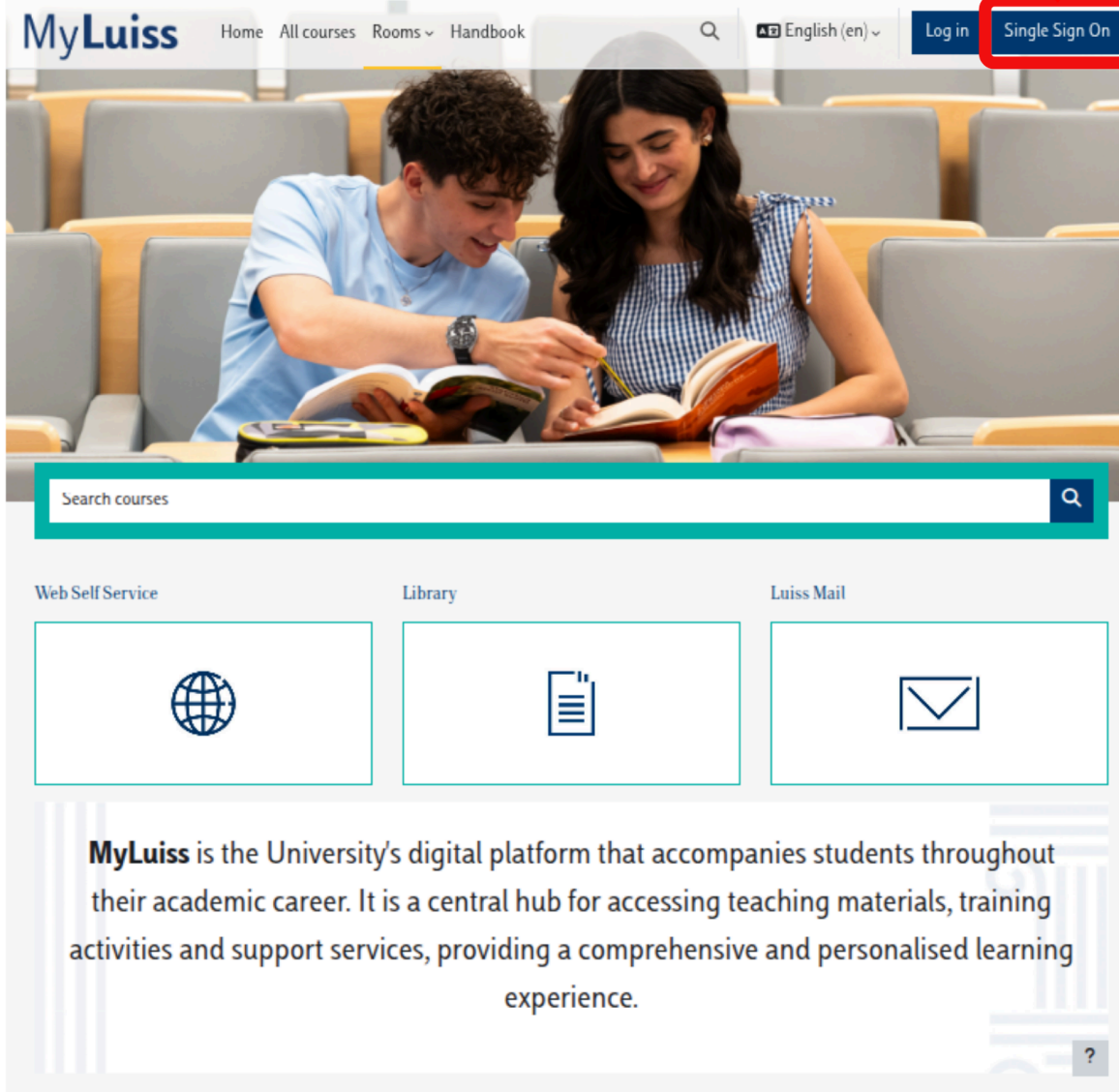
MyLUISS is our e-Learning platform: ***It is important that you get access!***

You'll find there:

- **Slides** covering:
 - theory
 - exercises
- **Details** on exam dates, reservations, results (besides standard Luiss channels)
- **Communications** from teacher and teaching assistants

A live and editable version of the material course is available at <https://introcp.github.io/>

CLICK HERE



The image shows the MyLuiss website interface. At the top, there is a navigation bar with the MyLuiss logo, links for Home, All courses, Rooms, and Handbook, a search icon, a language selector set to English (en), and buttons for Log in and Single Sign On. The Single Sign On button is highlighted with a red box and a red arrow pointing to it from the text 'CLICK HERE'. Below the navigation bar is a large banner image of two students studying in a lecture hall. Underneath the banner is a search bar labeled 'Search courses'. Below the search bar are three service tiles: 'Web Self Service' with a globe icon, 'Library' with a document icon, and 'Luiss Mail' with an envelope icon. At the bottom, there is a text block explaining that MyLuiss is the University's digital platform for students, serving as a central hub for teaching materials, training activities, and support services. A small question mark icon is visible in the bottom right corner of the page.

MyLuiss

Home All courses Rooms Handbook

English (en)

Log in Single Sign On

Search courses

Web Self Service

Library

Luiss Mail

MyLuiss is the University's digital platform that accompanies students throughout their academic career. It is a central hub for accessing teaching materials, training activities and support services, providing a comprehensive and personalised learning experience.

?

Course overview

Starred

Search

Sort by course name

Card



CLICK HERE



Logged in user



Emilio Coppa

Country: Italy

Email address: ecoppa@luiss.it
Courses
completions

0

Courses already
completed
Programs
Completions

0

Programs already
completed

Campus Life

[Biblioteca](#)[Tutorato](#)[Student Development](#)

Prima delle lezioni

[Consultare l'orario](#)[Attività preliminari \(Freshers'
Week, OFA, Precorsi\)](#)[Rilevazione delle presenze](#)

Upcoming events

There are no upcoming events
[Go to calendar...](#)



INTRODUCTION TO COMPUTER PROGRAMMING

[DASHBOARD](#) | [MY COURSES](#) | TDS03-N0-L25AIM-80726-2025[Course](#)[Information](#)[Grades](#)

Syllabus

[Collapse all](#) [Syllabus Attività Formativa](#) [Comunicazioni / Communications](#) [Contatti e orari di ricevimento / Contacts and office hours](#) [Software usati nel corso / Software used in the course](#) [Computer Systems](#) [Python](#) [Valutazione in itinere / In-course assessment](#) [Esame finale / Final Exam](#)

Access the *live* material

The course material will be made available on MyLUISS via slides. However, you can have access to:

- browsable content of the course
- editable notebooks with theory materials
- notebooks with exercises (easy to download or import on Google Colab and Jupyter Lite)

by accessing the <https://introcp.github.io/>



Search ctrl + K

Introduction to computer programming

Introduction

Course overview

Computer Systems

Bits and Bytes

Boolean Logic

Hardware

Software

Operating System Essentials

Python

Getting Started

Expressions and Data Types

Variables and Strings

Types of Errors

Branching

Functions

Iteration

Strings

Lists

Dictionaries, Tuples, and Sets

Classes and Objects



What is a Computing System?

A computing system:

- consists of **Software** and **Hardware** working together
- allows **program computations**, i.e., execution of programs



Common examples are your desktop PCs, laptops, smartphones, smartwatches, data centers, supercomputers, smart TVs, smart fridges, and so on!

Computations manipulate Data

Well, basically:

- **performs** calculations: a billion calculations per second!
- **remembers** results: hundreds of GigaBytes of storage!

What kind of **calculations**?

- *built-in* included in the programming language
- custom and ad-hoc defined by the programmer

The **essence** of a computation: store, manipulate, and move **data**.

Contents

What is a Computing System?

Computations manipulate Data

Data in computing means **bits**

Positional Notation

Why computers prefer the binary representation?

Humans and Computers are quite different

Binary representation

Conversion from binary to decimal

First n binary numbers ($n = 16$)

Interesting properties of binary numbers

Conversion from binary to decimal

Computers use hexadecimal numbers, why?

Hexadecimal numbers

Conversion from binary to hexadecimal

Conversion from hexadecimal to binary

Turning text into bits

Turning bits to text

Tool for the text-to-binary conversion

Text is for humans, binary is for machines

Other data formats

PPM Image format

Byte

Teachers



Emilio Coppa

- 6 out of 8 CFU
- Position: Assistant Professor Tenure-Track @ LUISS
- Research topics: software testing, vulnerability detection
- Extra: Team Manager of TeamItaly, the italian cybersecurity team
- Other info: <https://ecoppa.github.io>
- Office Hours: send me an e-mail so we can arrange a time slot.
- Contact: `ecoppa@luiss.it`



Alessio Martino

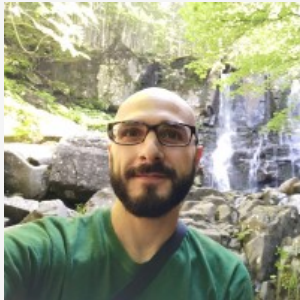
- 2 out of 8 CFU
- Position: Assistant Professor @ LUISS
- Research topics: machine learning, big data analysis, computational biology
- Office Hours: send me an e-mail so we can arrange a time slot.
- Contact: `amartino@luiss.it`

Teaching assistants



Giorgio Piccardo

- Position: PhD Student in Cybersecurity @ Sapienza-LUISS
- Research topics: large language models, cybersecurity, blockchain
- Other info: <https://giorgiopiccardo.com/>
- Office Hours: send me an e-mail so we can arrange a time slot.
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Ettore Di Micco

- M.Sc. in Electronic Engineering at Sapienza
- Research topics: Software Architecture and Design, Software Development and testing, Cloud Computing
- Office Hours: send me an e-mail so we can arrange a time slot.
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Course Goals

- The course is intended to teach **basic programming concepts** to students with no prior coding experience, developing an attitude towards **computational thinking**.
- It will provide the students with:
 - an understanding of the role that computation can play in solving problems
 - the ability to write small programs to accomplish useful goals

Course Roadmap

The course will familiarise the students with computer programming and will cover the following topics:

- Introduction to computer systems
- Binary numbers and boolean logic
- Computer hardware
- Computer software
- Operating Systems essentials
- Introduction to programming languages
- Python basics
- Variables, data types, expressions
- Control structures
- Functions and modules
- Recursion
- Strings
- Lists
- Dictionaries
- Object-oriented programming and classes

Organization

Lectures + Lab Sessions: *Laptops at your fingertips, always!*

- **Monday** @ 14:45-16:15 (90 minutes)
TD2a, TD2b @ Viale Romania
(often) Theory lectures & some exercises
- **Tuesday** @ 11:45-13:15 (90 minutes)
TD2a, TD2b @ Viale Romania
(often) Theory lectures & some exercises
- **Wednesday** @ 14:30-15:45
AT01 @ Viale Romania
(often) Lab Session with TA
- **Friday** @ 19:15 - 20:30 (75 minutes)
TD0 @ Viale Romania
(often) Lab Session with TA

Exam

Written exam:

- **Coding** (50% of the overall grade)
- **Multiple-choice Tests** (50% of the overall grade)

In light of the LUISS Continuous Assessment verification method, we will be having 2 *in itinere* exams:

- First Intermediate Test (week 7: 20-24 October)
 - 60 minutes, 8 theory quizzes, 2 coding exercises
- Second Intermediate Test (week 13: 1-5 December)
 - 75 minutes, 7 theory quizzes, 3 coding exercises

Each intermediate test is a short exam on the topics covered so far. It will consist of a mixture of coding exercises and theory questions (multiple-choice quiz). Note that:

- the sum of the coding parts between the two tests yields 31 points
- the sum of the theory parts between the two tests yields 31 points

NOTE: you must be a ***compliant student*** (at least 70% of attendance) to take the intermediate tests.

Exam scenario A: compliant student passing both tests

A compliant student taking the two intermediate tests has passed the exam if:

- the sum of the coding parts from the two tests is greater than 18
- the sum of the theory parts from the two tests is greater than 18

The final exam score is computed as the average of two scores.

The student can:

- accept the score:
 - GREAT!
 - **the student has to book in the first exam date**
 - there is no need to take the exam during the exam date!
- reject the score of the second test: see *Exam scenario B*
- reject the score at both tests: see *Exam scenario C*

Exam scenario B: reduced exam

A compliant student failing (too low score) or missing the second test can take a **reduced** exam (which is similar to the second intermediate test: 75 minutes, 7 theory quizzes, 3 coding exercises) **during the first exam date**.

The student passes the exam if:

- the sum of the coding parts from the first test and the reduced exam is greater than 18
- the sum of the theory parts from the first test and the reduced exam is greater than 18

The final exam score is computed as the average of these two scores.

The student cannot reject the final exam score if it is equal to or greater than 18.

Students failing the exam: see *Exam scenario C*

Exam scenario C: other situations

Students that:

- **are not compliant**
- **are taking the exam after the first exam date**
- are compliant but missed both tests
- have taken the tests but failed both tests
- have rejected the score of both tests
- have attempted the reduce exam (scenario B) and failed it

have to take the **full** exam (120 minutes):

- 15 theory quizzes (20 minutes)
- 5 coding exercises (100 minutes)

The final exam score is computed as the average of two parts (theory and coding).

Students cannot reject the score if it is equal to or greater than 18.

